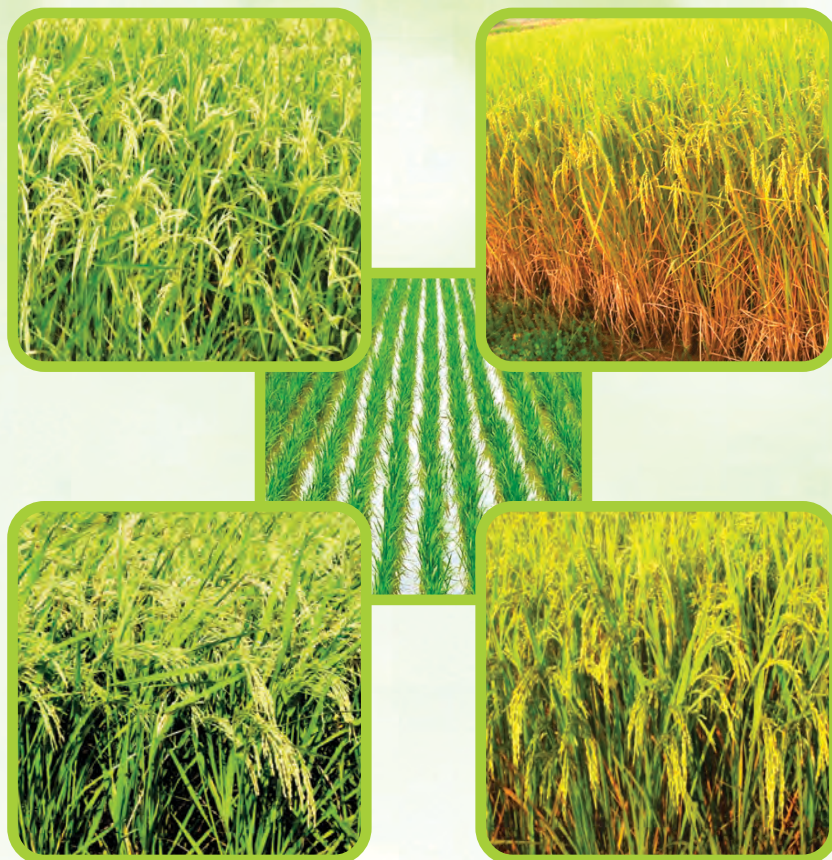


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A Anandan, R Sah, J Meher, L Behera, SK Dash & LK Bose



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High yielding rice varieties for rainfed shallow lowland ecology

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Rice is the main crop in India. The crop is cultivated in about 43 mha of land of which 16 million ha area is under rainfed lowland ecology. Large rice area of about 22 mha which represents about 45% of rice areas is cultivated as rainfed crop. The productivity and production is very low from this ecology. About 92% of rainfed lowland areas are located in the eastern region of the country. Depending upon the water depth and duration of water stagnation in the lowland, the lowland rice is categorized into rainfed shallow lowland, semi-deep, deep and very deep water or floating type ecosystem. The rainfed shallow lowland is characterized by water accumulation of about 0-50 cm that faces frequent drought and flash flood. Large area of rice is existing under shallow lowlands in eastern India with low production. The Green Revolution had no impact on the rainfed shallow lowland ecology of eastern India. This neglected ecosystem has now great potential to increase the yield and total production.

Shallow lowland faces frequent droughts and flash floods in single or different cropping seasons. This ecology is not fit for raising other important crops and therefore suitable for growing lowland paddy only. Climate resilient rice varieties. In future, production of rice needs to be increased even from stressful environments. In fact, with the increasing scarcity of water for rice cultivation, the future thrust for growing rice must focus on lowland ecologies. Rice crop faces several challenges when grown under shallow lowland ecology. Many rice growers under rainfed shallow lowlands adopt direct seeding of rice. Usually the direct seeded fields are not properly leveled before seed sowing and in case of severe rain water accumulates in the field which results rotting of germinating seeds and reduces plant population and hence produces low yield. Presence of anaerobic germination in the varieties is thus needed for direct seeded rice. Rice field submergence condition for more than 10 days is very common in most areas under this ecology. Though the ecology possess sufficient amount of water and stagnant water, but occasional drought spells also occur. Strong culm is needed in the varieties of this region due to occurrence of frequent high wind, frequent incidences of untimely rainfall during grain maturity which

causes heavy crop loss due to lodging and pre-harvest germination of seeds. Biotic stresses such as stem borer, brown plant hopper, leaf folder, gall midge, bacterial blight, false smut and bacterial sheath blight are severely affecting rice production. Few suitable varieties for different states for the shallow lowland ecology are presented in the following Table.

Table 1. Suitable rice varieties for the rainfed shallow lowland ecology

Sl. No.	Name of the varieties	Duration (days)	Recommended state	Avg. yield (t/ha)	Remark
1	Swarna-Sub 1	145	Odisha, UP	5.2	Submergence tolerance for 14 days
2	Ran-jit-Sub1	155	Assam	4.8	Submergence tolerance for 2 weeks
3	Bahadur-Sub1	155	Assam	5.0	Submergence tolerance for 2 weeks
4	CR Dhan 800	140	Odisha	5.75	Moderately resistant to bacterial leaf blight and sheath blight
5	CR Dhan 801	140	Odisha, West Bengal, Andhra Pradesh, Telangana, UP	6.3	Moderately resistant to leaf blast, neck blast, bacterial blight, stem borer (both dead heart and white ear heads) and leaf folder
6	CR Dhan 802 (Subhas)	142	Bihar, MP	6.5	Resistant to stem borer (both dead heart and white ear heads), leaf folder, plant hopper and case worm while moderately resistant to bacterial blight, sheath rot and rice tungro virus
7	CR Dhan 409 (Pradhandhan)	160-165	Odisha	5.0	LS grain, height:130-140 cm, moderately resistant to blast, neck blast, sheath blight, sheath rot, stem borer and leaf folder
8	Pooja	150	Orissa,, MP, Assam, Tripura, WB	4.5	Resistant to bacterial blight, Field tol. to major pests and diseases
9	CR Dhan 410 (Mahamani)	160-165	Odisha	7	Moderately resistant to stem borer, leaf folder, neck blast, bacterial blight, sheath rot and brown spot
10	Sumit	145	Odisha	5.2	Resistant to stem borer and leaf blight

11	CR Dhan 401 (Reeta)	150	Odisha, WB, AP, TN	5.5	Resistant to leaf blight, moderately resistant to 110 cm, R to Leaf blast; MR- Neck Blast, sheath blight and sheath rot
12	CR Dhan 408 (Chaka Akhi)	165	Odisha	4.8	Moderately resistant to neck blast, bacterial leaf blight, sheath rot, stem borer, leaf folder
13	CR Dhan 407	150	Odisha, WB	5	Moderately resistant to neck blast and leaf blight
14	Savitri	150-155	Odisha, TN	3.8	Moderately resistant to bacterial blight and sheath blight
15	Gayatri	155	Odisha	5.0	SB grain; Semi dwarf, MR to BB, MR-Blast and GM;
16	Luna Suvarna (CR Dhan 403)	150	Odisha	3.5-4.0	Suitable for saline shallow low-lands, R to blast, tolerant to brown plant hopper and sheath blight
17	CR Dhan 300	140	Odisha, Bihar, Maharashtra and Gujarat	5.0	LS grain, R to stem borer, leaf folder and rice whorl maggot, MR to white backed plant hopper, gallmidge bio-type 5, rice hispa and rice thrips.
17	Luna Sampad	140	Odisha	4.0	Resistant to blast, tolerant to SB, BPH, LF
18	CR Dhan 907 (aromatic)	150	Chhatisgarh, Odisha, Andhra Pradesh, Gujarat	4.5	MS grain, resistant to neck blast, gall midge, MR to sheath rot and stem borer
19	CR Dhan 908	145	Odisha, West Bengal and Uttar Pradesh	5.0	Height-95 cm, MR-Leaf blast, Neck Blast, Bacterial Leaf Blight, Brown Spot, Stem Borer, Leaf folder and WBPH.
20	CR Dhan 909	140	Assam, Bihar, UP, Maharastra	5.0	Moderately tolerant to Leaf blast, Neck Blast, Bacterial Leaf Blight, Brown Spot, Stem Borer, Leaf folder and WBPH.
21	CR Dhan 910	140	Odisha	4.4	Height-101 cm, MR to BL,NBl, ShR, RTV, SB, LF &WBPH

R: Resistant; MR: Moderately resistant; Bl: Blast; NBl: Neck blast; ShR: sheath rot; RTV: Rice tungro virus; SB: Stem borer; LF: Leaf folder; BB: Bacterial blight; BPH: Brown Plant hopper; WBPH: White backed planthopper;

Swarna-Sub1

The variety Swarna is cultivated in an estimated 25% area in the rain-fed lowlands. Swarna-Sub1 could almost replace Swarna variety. Swarna-sub-1 is a flood tolerant version of popular mega variety swarna (MTU7029) in eastern India. The plant height of Swarna sub 1 is 100 cm, short bold grain with green plant parts. About 140-145 days from seeding to maturity. It is tolerant to submergence and also suitable for late planting with aged seedlings. It has average yield of 5000 – 5500 kg/ha. It is suitable for shallow lowland areas of eastern India where Swarna was growing. Marker-assisted Selection approach was used to introgress the *Sub1* QTL into the Swarna variety. This variety can be recommended to the flood prone areas of eastern India where flash flooding is common and Sub1 varieties has potential of covering > 10 mha cultivated rice area. Swarna-Sub1 rice is genetically identical to Swarna except for the presence of a gene increasing flood tolerance.

Features of the variety, Swarna-Sub1

1	Coleoptile colour	Green
2	Basal leaf sheath colour	Green
3	Leaf: intensity of green colour	Dark
4	Leaf pubescence of blade surface	Medium to Strong
5	Leaf auricle	Present
6	Leaf: collar	Present
7	Leaf: ligule	Present
8	Leaf: shape of ligule	Split
9	Leaf: colour of ligule	Green
10	Leaf: Length of blade in cm	42.52
11	Leaf: width of blade mm	12.48
12	Flag leaf: attitude of blade	Erect to Semi erect
13	Culm attitude	Semi erect
14	Panicle : awn	Absent
15	Secondary branching in the panicle	Present
16	Panicle: secondary branching	Strong

17	Panicle attitude of branches	Semi erect
18	Panicle exertion	Partly Exserted to Exserted
19	Leaf senescence	Late
20	Sterile lemma colour	Straw
21	Seed test weight	19.86g
22	Hulling (%)	79.00
23	Milling (%)	74.00
24	HRR (%)	66.00
25	Length(mm)	5.40
26	Breadth(mm)	2.13
27	Length/Breadth ratio	2.54
28	Grain type	Medium Slender
29	Alkali value	5.0
30	Kernel length aftercooking	9.40
31	Elongation ratio	1.74
32	Amylose (%)	23.72



Swarna variety at dough stage of the crop



Fig. Swarna Sub-1 variety after recede of flood

CR Dhan 409 (Pradhan Dhan)

CR Dhan 409 (Pradhan Dhan) is very good grain type rice variety for the shallow lowland ecology. The variety was released by the SVRC, Odisha during the year, 2016. The average yield of the variety in Odisha is around 5 t/ha. Maturity duration of the variety is 160-165 days with semi-tall plant type (130-140 cm). Maturity duration of the variety is 160-165 days with semi-tall plant type (130-135cm). It possesses long slender grain, more panicles per m² (250-300) with 135-140 days to 50% flowering, high tillering (12-15), medium and dense panicle with moderate test weight (22.5g). It is moderately resistant to leaf blast, neck blast, sheath blight, sheath rot, stem borer (both dead heart and white ear heads) and leaf folder. CR Dhan 409 has good hulling, milling and head rice recovery. It possesses intermediate amylose content, long slender grain and other desirable grain quality parameters. The base of the plant is purple in colour and hence suitable for separating weeds from rice seedlings in direct seeded plot. CR Dhan 409 is suitable for puffed rice, puffed paddy, beaten rice, wet rice preparations and fried rice also.



Fig. Field view of rice variety, CR Dhan 409 (Pradhan Dhan) at dough stage of the crop



Fig. Panicles of rice rice
Pradhan dhan

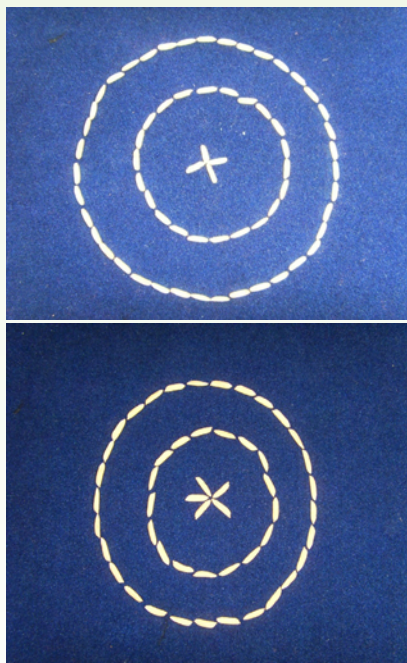


Fig.Kernels and grains of rice
variety Pradhan dhan
(CR Dhan 409)

CR Dhan 401 (Reeta)

CR Dhan 401 (Reeta) was released in the year 2010 by CVRC. The selection and evaluation work was carried out at National Rice Research Institute, Cuttack, Odisha, India. The variety is moderately tolerant to neck blast, brown spot, sheath blight, sheath rot, green leaf hopper, stem borer, leaf folder and gall midge biotype4. CR Dhan 401 has good hulling, milling and head rice recovery. Maturity duration of the culture is 145-150 days, semi-dwarf plant type, non-lodging genotype, suitable for shallow lowland / irrigated late maturity group in the states of Odisha, West Bengal, Tamil Nadu and Andhra Pradesh. It produces medium slender grain, 250-300 panicles per m² with 120-125 days to 50% flowering, moderate tillering (7-10), extra-long panicle (28-31 cm) with seed test weight of 30.5 gm. The variety produces 5.4 t/ha under rainfed shallow lowlands of India. The variety possesses desirable cooking and eating grain characteristics. The plant height of CR Dhan 401 is 110 cm. It is weakly photosensitive, Semi-dwarf (110 cm), long panicle, medium slender grain with white kernel, broad and upright flag leaf and thick leaf.

DESCRIPTION OF THE VARIETY:

1.	Plant height	110cm
2.	Plant type	Semi-dwarf
3.	Av. tillers/plants	8
4.	No. of panicles/ m ²	280
5.	Flowering duration	120 days
6.	Panicle type	Compact long
7.	Panicle exertion	Well exerted
8.	Awning	Awnless
9.	Apiculus colour	Straw
10.	1000-grain weight	30.5 g
11.	Kernel length (mm)	5.34mm
12.	Kernel breadth (mm)	1.97mm
13.	L/B ratio	2.7
14.	Grain type	MS
15.	Kernel appearance	White
16.	Milling recovery	72.6%
17.	Head - rice recovery	62.4 %
18.	Alkali value	4.0-5.0
19.	Amylose content	22.34 %
20.	Gel consistency	24



Fig. Kernels, panicles and grains of variety, CR Dhan 401

CR Dhan 408 (Chakaakhi)

CR Dhan 408 was released by SVRC, Odisha for the shallow lowlands of the state during the year 2014. The average yield of the cultivar in Odisha is 4.8 t/ha. The variety is strongly photosensitive and it flowers in the 1st week of November. The plant height is 130-140cm. It has long bold grain, 250-300 panicle per m², moderate tillering (7-10) and having seed test weight of 25.6g. The base of the variety is purple in color and seed is apiculous. The cultivar can tolerate one-week submergence tolerance. CR Dhan 408 possess desirable quality characters like high head rice (60%), high milling (68%), intermediate amylose content (23.03%) and good for wet rice preparation. The variety is moderately resistant against the stem borer, leaf folder, white backed plant hopper and rice whorl maggot attack. It is also moderately tolerant to Leaf blast, neck blast, bacterial blight and sheath rot diseases.

Morphological features of the variety, CR Dhan 408

1.	Plant height	130--140 cm
2.	Plant type	Semi-tall
3.	Av. tillers/plants	7-10
4.	No. of panicles/ m ²	250-300
5.	Flowering duration	November 1 st week
6.	Panicle type	Compact panicle
7.	Panicle exertion	Well exerted
8.	Awning	Sporadic
9.	Apiculus colour	Purple
10.	1000 -grain weight	25.5 g
11.	Kernel length (mm)	6.1 mm
12.	Kernel breadth (mm)	2.46 mm
13.	L/B ratio	2.48
14.	Kernel appearance	White
15.	Grain type	Long bold
16.	Milling recovery	68 %
17.	Head - rice recovery	59%
18.	Alkali value	4.0
19.	Amylose content	23.03 %
20.	Aroma	Absent



Fig. Photographs showing field view, panicles, kernels and grains of CR Dhan 408

CR Dhan 404 (Sumit)

CR Dhan 404 (Sumit) recommended for the rainfed shallow lowlands of Odisha. CR Dhan 404 was released by SVRC, Odisha in the year 2016. Its maturity duration is 150 days, semi-dwarf (108-115 cm), non-lodging genotype, suitable for shallow lowland/irrigated late maturing rice growing areas of the state. It produces long bold grain, 260-300 panicles per m² with 120-125 days to 50% flowering, moderate tillering (6-8), extra-long and heavy panicle (29-31 cm). The variety is resistant to leaf blast, stem borer, leaf folder, glume discoloration and GRH. It is resistant to both dead heart and white ear head stem borer insects. It is moderately tolerant to neck blast, brown spot, bacterial leaf blight, sheath blight, sheath rot, green leaf hopper, gall midge biotypes 1, 5, 6, rice thrips and EHB. It tolerates flash flood submergence for one week and has good regeneration ability after natural de-submergence. It has 6.26 mm of grain length, 2.45 mm of kernel length, 3.68 L/B ratio, 70.3% of milling recovery, 62.2% of head-rice recovery, 4.0-5.0 alkali value and 24.03% amylose content. The average yield of the variety is 5.2 t/ha.

Description of the Variety: Sumit (IET 19913)

1.	Plant height	112cm
2.	Plant type	Semi-dwarf
3.	No. of tillers/plants	9

4.	No. of panicles/ m ²	275
5.	Flowering duration	122 days
6.	Panicle type	Compact long panicle
7.	Panicle exertion	Well exerted
8.	Awning	Awnless
9.	Apiculus colour	Normal
10.	1000-grain weight	33.5 g
11.	Kernel length (mm)	6.26 mm
12.	Kernel breadth (mm)	2.45 mm
13.	L/B ratio	3.68
14.	Grain type	LB
15.	Kernel appearance	White
16.	Milling recovery	70.3%
17.	Head - rice recovery	62.2%
18.	Alkali value	4.0-5.0
19.	Amylose content	24.03 %
20.	Gel consistency	47.5



Fig. Field view of rice variety, CR Dhan 404 (Sumit)

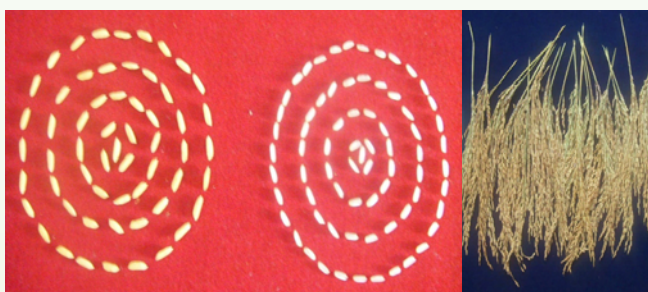


Fig. Panicles and kernels of rice variety, CR Dhan 404 (Sumit)

CR Dhan 800

CR Dhan 800 is released for Odisha state by SVRC, Odisha during the year 2016. The variety is resistant to bacterial blight of rice. CR Dhan 800 was developed by pyramiding of *xa5*, *xa13* and *Xa21* bacterial blight resistance genes in the popular variety, Swarna. It is developed through marker-assisted breeding approach at the National Rice Research Institute, Cuttack. This variety has the medium slender grain, intermediate alkali spreading value and amylose content identical to the parent, Swarna. The variety is suitable for bacterial blight endemic areas Odisha state.

DESCRIPTION OF THE VARIETY, CR Dhan 800

1.	Plant height	85-90 cm
2.	Plant type	Semi-dwarf
3.	No. of tillers/plant	9-12
4.	No. of panicles/m ²	250-300
5.	Flowering duration	115 days
6.	Panicle type	Compact panicle
7.	Panicle exertion	Well exerted
8.	Awning	Awnless
9.	Apiculus colour	Green
10.	1000 -grain weight	18.98 g
11.	Kernel length (mm)	5.30 mm
12.	Kernel breadth (mm)	2.06 mm
13.	L/B ratio	2.57
14.	Kernel appearance	White
15.	Hulling recovery	79.6 %
16.	16. Milling recovery	69.9 %
17.	Head - rice recovery	66.7 %
18.	Alkali value	4-5
19.	Amylose content	22.68 %



Fig. Photographs showing field view, panicles, kernel and grains of CR Dhan 800

CR Dhan 801

The variety, CR Dhan 801 is recommended for cultivation in the states of Odisha, West Bengal, Andhra Pradesh, Telangana and Uttar Pradesh. It was developed from the cross of IR81896-B-B-195/2*Swarna-Sub1//IR91659-54-35 by marker-assisted breeding method. The variety has tolerance to submergence and drought condition. The variety is weakly photosensitive with average maturity duration of 140-145 days with plant height of 100cm. The average yield of the variety is 6350 kg/ha under normal condition and about 3000 kg/ha under drought and submergence stress condition. It possesses short bold grain with a test weight of 20.5g. It is resistant to stem borer (both dead heart and white ear heads), leaf folder and case worm while moderately resistant to bacterial blight and rice tungro virus. It has good hulling, milling, head rice recovery, alkali spreading value and intermediate amylose content. It possesses intermediate amylose content, short bold grain and other desirable grain quality parameters.

Characteristics of variety, CR Dhan 801

Sl. No.	Plant height	100 cm
1	Flowering	114 days

2	Maturity duration	140 days
3	1000-grain weight (g.)	20.6
4	Lodging	Non-lodging
5	Panicle type	Intermediate
6	Panicle exertion	Well exeserted
7	Hulling (%)	79.6
8	Milling (%)	69.9
9	Head Rice Recovery (%)	66.2
10	Kernel Length(mm)	5.15
11	Kernel Breadth(mm)	2.22
12	L/B Ratio	2.31
13	Grain type	SB
14	Grain Chalkiness.	VOC
15	Alkali Spreading Value	4.0
16	Amylose Content (%)	25.13
17	Gel Consistency (mm)	39



Fig. Field view of CR Dhan 801 at milking stage of the crop



Fig. Panicles and kernels of variety, CR Dhan 801

CR Dhan 802 (Subhas)

The variety, CR Dhan 802 (Subhas) was developed from the breeding materials of cross Swarna-Sub1 *4 / IR81896-B-B-195 is recommended for cultivation in the states of Bihar and Madhya Pradesh. Selection, evaluation and testing of the NILs were done at ICAR-National Rice Research Institute, Cuttack, Odisha with the objective to breed high yielding variety with tolerance to submergence and drought condition. CR Dhan 802 was released in the year 2019. The average yield of the variety is 6508 kg/ha under normal condition and 2344 kg/ha under drought stress. CR Dhan 802 is weakly photosensitive with average maturity duration of 140-145 days. It possesses short bold grain with a test weight of 19g. It is resistant to stem borer (both dead heart and white ear heads), leaf folder, plant hopper and case worm while moderately resistant to bacterial blight, sheath rot and rice tungro virus. It possesses intermediate amylose content, short bold grain and other desirable grain quality parameters.

Characteristics of variety, CR Dhan 802 (Subhash)

Sl.No.	Plant height	100 cm
1	Flowering	116 days
2	Maturity duration	145 days
3	1000-grain weight (g.)	20.4
4	Lodging	Non-lodging
5	Panicle type	Intermediate
6	Panicle exertion	Well exeserted
7	Hulling (%)	77.85
8	Milling (%)	70.2
9	Head Rice Recovery (%)	64.25
10	Kernel Length (mm)	5.0
11	Kernel Breadth (mm)	2.16
12	L/B Ratio	2.31
13	Grain type	SB
14	Grain Chalkiness.	VOC
15	Alkali Spreading Value	4.5
16	Amylose Content (%)	25.0
17	Gel Consistency (mm)	39.5



Fig. Field view of rice variety CR Dhan 802



Fig. Photographs showing panicles and kernels of CR Dhan 802

CR Dhan 410 (Mahamani)

The variety CR Dhan 410 (Mahamani) is a premium quality shallow lowland variety released by SVRC, Odisha in the year 2019. The mean yield in Odisha state is found to be 4353 kg/ha. The genotype is strongly photosensitive with average maturity duration of 155 days. It possesses long slender grain with a long heavy panicle having moderate test weight (24g). It is resistant to stem borer (both dead heart and white ear heads) and leaf folder while moderately resistant to neck blast, bacterial blight, sheath rot and brown spot diseases. CR Dhan 410 has good hulling, milling and head rice recovery. It possesses intermediate amylose content, long slender grain and other desirable grain quality parameters.

Characteristics of variety, CR Dhan 410 (Mahamani)

Sl.No.	Plant height	140-150 cm
1	Flowering	131 days
2	Maturity duration	155-158 days
3	1000-grain weight (g.)	24.0
4	Lodging	Non-lodging

5	Panicle type	Intermediate
6	Panicle exertion	Well exserted
7	Hulling (%)	79.2
8	Milling (%)	69.8
9	Head Rice Recovery (%)	67.5
10	Kernel Length (mm)	6.22
11	Kernel Breadth (mm)	2.25
12	L/B Ratio	2.77
13	Grain type	LS
14	Grain Chalkiness.	A/VOC
15	Alkali Spreading Value	4.0
16	Amylose Content (%)	23.96
17	Gel Consistency (mm)	22.0



Fig. Field view of CR Dhan 410 (Mahamani) at dough stage of the crop



Fig. Photographs of grains and kernels of variety, CR Dhan 410

CR DHAN 300

The super fine variety CR Dhan 300 is recommended for release in the states of Maharashtra, Gujarat, Odisha and Bihar. The average yield of the entry in region-IV is about 5.4t/ha. Days to 50% flowering of the culture in these region is 113 days, semi-dwarf (100-105 cm), non -lodging variety. It produces long slender grain (7.21 mm), 287 panicle per m², good tillering (9-13) with long panicle (28 cm). CR Dhan 300 possess all desirable quality characters like high head rice (65.7%), intermediate alkali spreading value (5.0), intermediate amylose content (24.8%) and volume expansion ratio of 4.3. The variety resistant to the pests stem borer, leaf folder and rice whorl maggot attack. While it showed moderate reaction to white backed plant hopper, gallmidge biotype5, rice hispa and rice thrips. It exhibited moderately tolerance reaction to diseases like Leaf blast, neck blast, sheath rot and rice tungro virus disease.

DESCRIPTION OF THE VARIETY: CR Dhan 300 (IET 19816)

1.	Plant height	105-110 cm
2.	Plant type	Semi-dwarf
3.	No. of tillers/plants	9-12
4.	No. of panicles/ m ²	300-350
5.	Flowering duration	113 days
6.	Panicle type	Compact panicle
7.	Panicle exertion	Well exerted
8.	Awning	Awnless
9.	Apiculus colour	Green
10.	1000 -grain weight	24 g
11.	Kernel length (mm)	7.21 mm
12.	Kernel breadth (mm)	1.94 mm
13.	L/B ratio	3.5 mm
14.	Kernel appearance	White
15.	Hulling recovery	79.6 %
16.	Milling recovery	72.0 %
17.	Head - rice recovery	66.6 %
18.	Alkali value	5.0
19.	Amylose content	24.99 %
20.	Aroma	Absent



Fig. Field view of rice variety, CR Dhan 300 at dough stage



Fig. Grain and kernels of rice variety, CR Dhan 300

CR Dhan 407

The CR Dhan 407 is recommended for the rainfed shallow lowland ecosystem of the states of Odisha and West Bengal. It has plant height of 102 cm, duration of 150 days, and well-exserted compact panicles, with average yield of around 5 t ha⁻¹. It showed moderate resistance to bacterial blight, leaf blast, white backed plant hopper and whorl maggot. It produces desirable grain quality features with long grain, high head rice recovery-HRR (63.6%), intermediate amylose content (23.475%), and intermediate gel consistency (GC).

Pooja (CR-629-256)

It is a late maturing (150 days) short statured (90-95 cm) popular variety, released and notified (1999) for cultivation in the shallow lowland of Odisha, Assam, Madhya Pradesh and West Bengal. It produces medium slender grains with an average yield of 5.0 t/ha. It possesses field tolerance to all major diseases and pests. It tolerates water stagnation (up to 25cm) and is suitable for late transplanting with aged seedlings.



Fig. Field view of rice variety, Pooja at milking stage

CR1009-Sub1

CR1009- Sub1, a new version of Savitri variety tolerant to submergence for 2 weeks. The variety is developed through marker-assisted selection from the cross CR1009 / FR 13 A. CR1009-Sub1 rice variety produces average grain yield of 5759 kg/ha. The variety is moderately resistant to BPH, WBPH, Brown spot and Leaf Blast. It has short bold grain with good milling percentage (69.8%) and head rice recovery (62.5%), intermediate gelatinization temperature, soft gel consistency and high amylose content (25.0), retaining the traits of Savitri variety.

Gayatri (CR 210-1018)

It is a late duration (160 days), semi tall (110 cm) photosensitive popular variety, released and notified (1988) for cultivation in low land of state Odisha, West Bengal and Bihar. It has short bold good cooking quality grains with average productivity of 5.0 t/ha. It has field tolerance against major diseases and pests. It has significant extent of grain dormancy, can tolerates up to 50 cm water stagnation and suitable for delayed transplanting.



CR Sugandh Dhan 907 (CR 2616-3-3-31)

It is a late maturing (150 days), first dwarf aromatic (non basmati) rice variety, recently released and notified (2012 and 2013) for cultivation in irrigated late ecology of Chhattisgarh, Odisha, Andhra Pradesh and Gujrat. It has medium slender grains with average productivity of 4.5-5.0 t/ha. It is resistant to neck blast, gall midge and moderately resistant to sheath rot and yellow stem borer.



Nua Dhusara

It is a late maturing (145 days), tall (142cm) and photosensitive popular variety, released and notified (2008) for cultivation in low land area of Odisha. It has short bold grains with average productivity of 3.0 t/ha. It is resistant against sheath rot, neck blast and RTV; moderately resistant against gall midge. It can also withstand with short period (3-4 days) and suitable for organic system of farming.



Nua Chinikamini

It is a late maturing (145-150 days), tall (140cm), photosensitive, aromatic (non-basmati) variety, released and notified (2010 and 2011) for cultivation

in lowland and rainfed low land area of Odisha. It has short bold grains with average productivity of 3.5 t/ha. It is resistant to sheath rot, neck blast and RTV and gall midge; moderately resistant to stem borer. It is recommended for close planting.



CR Dhan 909

CR Dhan 909 (CRL74-89-2-4-1) was developed from the cross Pankaj/ Padumoni by pedigree method of selection. The variety has been recommended for cultivation in the states of Bihar, Assam and Maharashtra. The mean grain yield of the variety was 4453 kg/ha. Plant height of the variety is 105-110 cm with late maturity duration. The variety is tolerant to leaf blast, neck blast, sheath rot and RTD. CR Dhan 909 was moderately resistant against stem borer (dead heart as well as white ear head), leaf folder and whorl maggot under natural screening. The variety showed excellent quality parameters like very high Head Rice Recovery of 70.0%, intermediate amylose content (24.02%) with medium GC (47 mm). It has medium slender grains with strong aroma and alkali spreading value (ASV) of 7.0.

Data on other important characters of the variety CR Dhan 909

1	Plant height	105-110 cm
2	Plant type	Semi-dwarf
3	No. of Tillers/plant	8-12
4	No. of panicles/ m ²	250-350
5	Days to 50% flowering	111-114
6	Panicle type	Compact
7	Panicle exertion	Well exerted

8	Awning	Absent
9	Apiculus colour	Normal
10	1000-grain weight (g)	24.6
11	Kernel length (mm)	5.80
12	Kernel breadth (mm)	2.19
13	L/B ratio	2.65
14	Kernel appearance	Translucent
15	Hulling recovery	79.1%
16	Milling recovery	71.3%
17	Head rice recovery	70.0%
18	Alkali spreading value	7.0
19	Amylose content	24.02
20	Aroma	Strongly scented



Fig. Grains and kernels of rice variety, CR Dhan 909

CR Sugandh Dhan 910

The variety CR Sugandh Dhan 910 was developed from the cross of Swarna and Geetanjali. This is an aromatic long slender grained variety with pleasant aroma. The variety was released in the year 2015 by SVRC in Odisha. The average grain yield of the variety was 4385 kg/ha. Maturity duration of the proposed variety is 142-145 days and the plant type is semi-dwarf. The proposed variety has panicles 263-277/ m². It is moderately resistant to blast, neck blast, sheath rot, rice tungro virus, stem Borer, leaf folder and white backed planthopper. The medium slender grains of the variety has a test weight of 17.3g with no kernel chalkiness, has high HRR (68.8%). It possesses soft GC, intermediate amylose content and ASV. The proposed variety with its high yield combined with superior quality aromatic rice, will fetch a premium market value to the farmers.



Fig. Photographs showing field view of CR Sugandha Dhan 910



Fig. Panicles, grains, Milled rice and Brown rice of CR Sugandh Dhan 910

CR Dhan 403 (Luna Suvarna)

CR Dhan 403 (Luna Suvarna) is released by SVRC, Odisha in the year 2010. It is a salt tolerant, tall, long duration rice variety suitable for coastal saline areas with resistant to major pest and diseases. The plant height of Luna Suvarna is 130 cm. It is photosensitive, medium tall, dense having fully exerted long panicles. It is a medium slender golden colour grain. The days to maturity of Luna Suvarna is 145-150 days. It has resistance against blast and moderate resistance against sheath blight and brown spot disease. The variety has tolerance to salinity (5.0 to 8.0 dsM⁻¹) and standing water upto 45cm depth. The average yield is about 3500-4000 kg/ha. It has resistance against blast and sheath blight, and moderate resistance for brown spot.

Grain quality characteristics of the variety

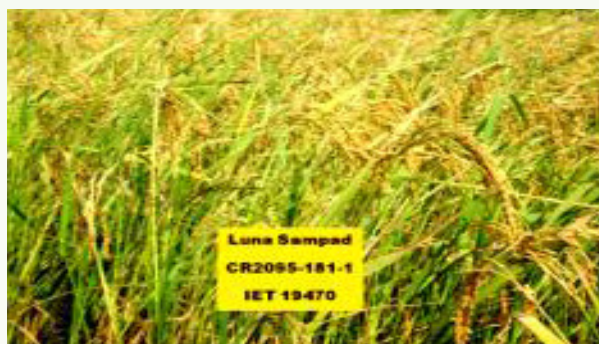
Quality characters	Luna Suvarna
Mill (%)	66.4
HRR (%)	62.9
KL (mm)	6.21
KB (mm)	2.22
L/B ratio	2.79
Grain Type	MS
Grain Chalk	A
ASV	5.0
AC (%)	25.08
GC (mm)	22



Fig. Grains and panicles of variety, CR Dhan 403 (Luna Suvarna)

Luna Sampad (CR2095-181-1)

It is a medium late maturing (140 days), tall (130 cm) saline tolerant (5.0 to 8.0 ds M⁻¹) variety, released and notified (2010 and 2011) for cultivation in coastal saline/rainfed saline situation of Odisha. It has medium bold grains with an average productivity of 3.6-4.2 t/ha. It has resistance ability against blast; moderately resistant to rice tungro virus, sheath blight and stem borer etc.



Luna Barial (CR 2092-1-3)

It is a late duration (150-155 days) saline tolerant (5.0 to 8.0 ds M⁻¹) variety, released and notified (2012 and 2012) for cultivation in coastal saline area of Odisha. It has short bold grains with an average productivity of 3.9 t/ha. It is resistant to yellow stem borer and leaf folder; moderately resistant to sheath blight and bacterial leaf blight etc.





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